

HearthGen: Hearthstone Card Artwork Generation with Knowledge Graphs and Diffusion Models

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Abstract

*Generating Hearthstone-style DIY cards requires both structured card semantics and visually consistent artwork. In this project, we present **HearthGen**, an end-to-end system that starts from raw card data and builds two connected branches: a LoRA training branch based on cropped artwork and structured semantics, and a knowledge-graph branch based on further entity and relation extraction for reference retrieval. During inference, the system takes a user description as input, generates structured semantics, retrieves relevant reference cards, and uses a diffusion model to generate Hearthstone-style artwork. Finally, the generated artwork is combined with the card description and other structured card elements to compose the final DIY card.*

1. Introduction

Hearthstone is a popular digital collectible card game with more than 6,000 collectible cards. Each card combines several types of information, including structured attributes such as card type, class, tribe, and card text, together with a central artwork image.

In this project, we study how to build a system that supports Hearthstone DIY card generation from raw card data to final card composition, with a particular focus on generating the artwork region.

Our main contributions are as follows:

- We build an end-to-end pipeline that starts from raw card data and connects preprocessing, structured semantic generation, knowledge graph retrieval, artwork generation, and final card composition.
- We design a knowledge graph construction pipeline that further extracts entities and relations from structured semantics to support reference card retrieval during generation.
- We build a training data construction pipeline that pairs cropped artwork with structured semantics to form a



Figure 1. Left: a screenshot of Hearthstone gameplay. Right: an example Hearthstone card. (will fix alignment in final version)

LoRA fine-tuning dataset for adapting the diffusion model to the Hearthstone visual domain.

2. System Architecture

Figure 2 shows the overall architecture of HearthGen. Starting from raw Hearthstone card data, the system first preprocesses card texts and images to obtain structured semantics and cropped artwork. These outputs are then used along two connected branches: one branch constructs a knowledge graph for semantic retrieval, while the other branch builds a LoRA fine-tuning dataset for diffusion model adaptation. During inference, the knowledge graph and the tuned diffusion model are combined to support the generation of new DIY cards.

2.1. Data Preprocessing

We start from raw Hearthstone card data, including card JSON records and original card images. For each card, we extract the basic structured fields needed for downstream processing, such as card type, class, tribe, card text, and other available metadata. In parallel, we preprocess the card images to isolate the central artwork region.

The image preprocessing step is designed to remove non-artwork elements such as the card frame, icons, and text box. We apply card-type-specific masks to the original card images, crop the valid artwork region, and resize the results

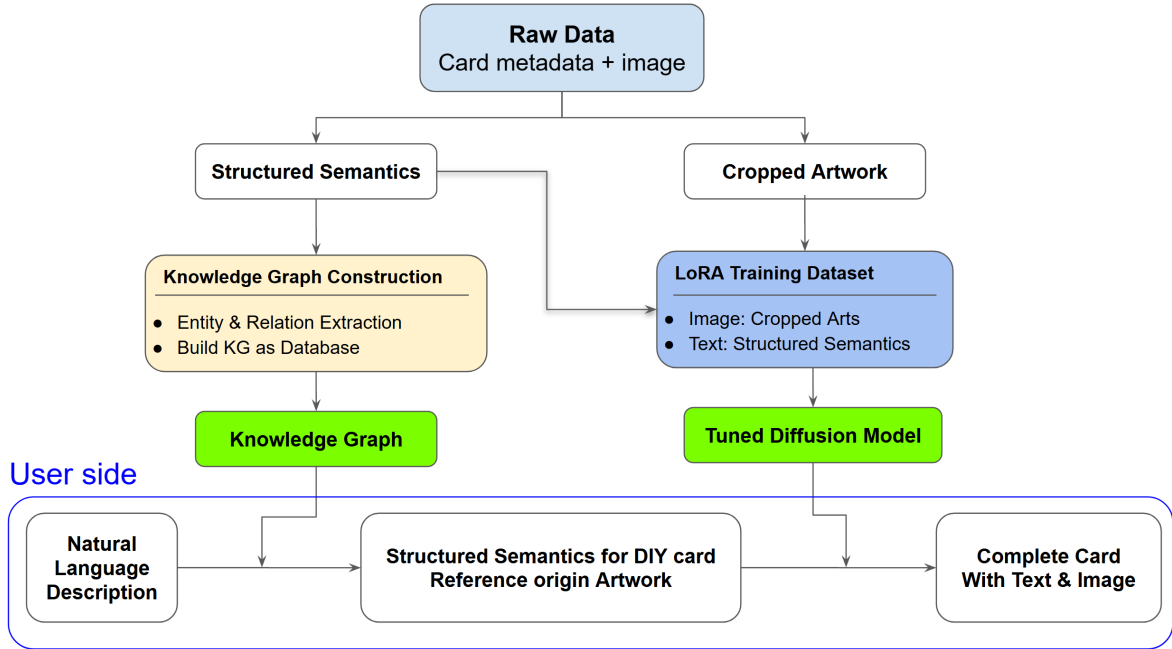


Figure 2. Overview of the HearthGen system. Raw card data is preprocessed into structured semantics and cropped artwork. These outputs are used to construct a knowledge graph and a LoRA training dataset, which support the user-side workflow for structured card generation, reference retrieval, artwork synthesis, and final card composition.



Figure 3. Data preprocessing pipeline. From left to right: the original card image, the mask used for artwork extraction, and the cropped artwork result.

to a unified resolution. This step also handles practical issues such as inconsistent image resolutions and imperfect alignment between the original card image and the mask template. The final output of this stage consists of two parts: structured semantics extracted from card data, and cropped artwork images used in later training and generation stages.

2.2. Knowledge Graph Construction

The structured semantics produced in the preprocessing stage are further used to build a knowledge graph over real Hearthstone cards. In this stage, we apply an LLM-based extraction step to identify entities, mechanics, and relations from the structured card representation, and then organize them into a graph structure.

```

{
  "card_name": "Al'Akir the Windlord",
  "card_type": "minion",
  "class": "Shaman",
  "tribe": "Elemental",
  "keywords": ["Charge", "Divine Shield", "Taunt", "Windfury"],
  "description": "Charge, Divine Shield, Taunt, Windfury.",
  "style": ["elemental", "storm", "lightning", "blue energy", "hearthstone-style"]
}

```

Figure 4. Example of structured semantics derived from a real Hearthstone card. This representation is later used for both knowledge graph construction and diffusion-side conditioning.

The resulting graph serves as an external knowledge source for later generation. Instead of directly generating from free-form user input alone, the system can use this graph to retrieve reference cards with related semantics and mechanics. In this way, the knowledge graph provides structured support for card design and improves the relevance of the conditioning information used during generation.

Figure 4 shows an example of the structured semantics extracted from a real card. Based on representations like this, we further identify shared entities and mechanics across cards and organize them into the local knowledge graph, as illustrated in Figure 5.

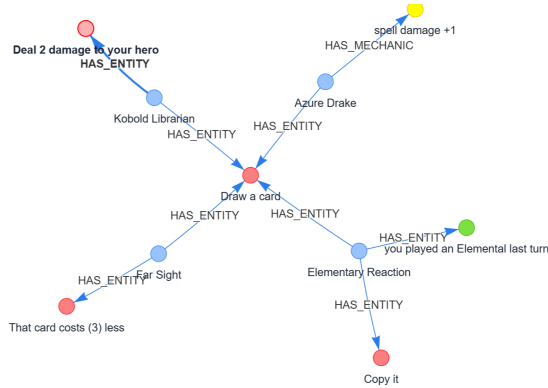


Figure 5. A local cluster from the knowledge graph.

2.3. LoRA Training Data Construction

In parallel with graph construction, we build a training dataset for diffusion model adaptation. Each training example pairs a cropped artwork image with its corresponding structured semantic description. This dataset is used to fine-tune a Stable Diffusion model with LoRA, so that the generator can better match the visual style of Hearthstone artwork.

This branch focuses on visual adaptation rather than game logic. The purpose of LoRA fine-tuning is to improve style consistency, composition, and domain-specific appearance, while keeping the training cost relatively low.

2.4. User-Side Generation Workflow

During inference, the system takes a user description of a DIY card as input. Using the knowledge graph built from real Hearthstone cards, it retrieves relevant reference cards and helps construct structured semantics for the target card. The structured semantics and reference card images are then used together as inputs to our LoRA-adapted Stable Diffusion model to generate a new artwork image.

Finally, the generated artwork is combined with the card description and other card elements to compose the final DIY card. In this way, the user-side workflow connects the structured support from the knowledge graph with the visual generation ability of the tuned diffusion model.

3. Current Progress and Preliminary Results

We have completed the main data processing pipeline, including raw card collection, structured semantic extraction, artwork cropping, and a local knowledge graph demo. These components already support a basic prototype workflow: structured semantics can be used for graph construction, and the resulting graph can provide reference cards for downstream generation.

On the generation side, we have run a Stable Diffusion baseline and tested reference-based conditioning. Our cur-

rent qualitative results show that the baseline model can generate fantasy-style images, but the outputs are often unstable in style. Adding reference images improves consistency and makes the results visually closer to Hearthstone artwork. The next step is to complete LoRA fine-tuning and evaluate its effect more systematically.

4. Evaluation Plan

In the final report, we plan to include both qualitative and quantitative evaluation. Qualitatively, we will compare text-only generation, text+reference generation, and LoRA-adapted generation, focusing on style consistency and semantic alignment. Quantitatively, we plan to use CLIP similarity for text-image alignment, embedding distance for visual style similarity, and a real-vs-generated classification setup to measure how close the generated artwork is to the real Hearthstone distribution.

These evaluations are not yet complete, but they define the main directions for the next stage of the project.

5. Future Work

Our next step is to scale both branches of the system. On the knowledge side, we plan to expand graph construction from the current prototype to the full collectible card set. On the generation side, we plan to finish LoRA fine-tuning on a larger artwork dataset and compare its effect against the current baseline. We will also complete the evaluation pipeline and improve the final card composition stage.